XUEXING ZHAO

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Seeking Summer 2027 Research Internship

EDUCATION

Oklahoma State University — Stillwater, OK
Ph.D. in Applied Mathematics — *Aug 2025 – Present (Expected 2030)*

- Advisor: Prof. Eduardo V. Teixeira
- Research Interests: Partial Differential Equations, Free Boundary Problems

Technical University of Munich — Munich, Germany
M.S. in Mathematics — *2021 – 2025*

- Master's Thesis: Random Dynamical Systems and the Fokker-Planck Equation
- Relevant Coursework: Stochastic Analysis, Deep Learning, Machine Learning

University of Shanghai for Science and Technology — Shanghai, China
B.S. in Mathematics — *2017 – 2021*

RESEARCH & INDUSTRY EXPERIENCE

Huawei Research Center Munich — Applied Network Technology Lab — Munich, Germany
Research Intern — *Sep 2024 – Mar 2025*

- Researched reinforcement learning (RL)-based knowledge distillation techniques for mathematical reasoning in large language models (LLMs), distilling reasoning capabilities from Qwen2.5-Math-7B-Instruct into smaller student models
- Designed a twisted Sequential Monte Carlo (SMC) sampling scheme, grounded in Feynman-Kac particle filtering theory, to generate multiple candidate reasoning traces per problem, assign each a posterior probability, and select the highest-probability candidate as the distillation training target
- Evaluated distillation into Qwen2.5-Math-1.5B as the student model, benchmarking retained mathematical reasoning ability against the Qwen teacher model on the GSM8K and MATH benchmarks

DARPA-Funded Math Formalization Project — OSU & NYU — Stillwater, OK / New York, NY
Research Assistant — *Mar 2026 – Aug 2026*

- Contributed to a joint OSU–NYU project funded by DARPA's Exponentiating Mathematics (expMath) program, formalizing mathematics in the Lean theorem prover
- Wrote Lean code to formalize and machine-verify results from Evans' Partial Differential Equations: the four fundamental linear PDEs (transport, Laplace, heat, wave) and the foundational theory of Sobolev spaces
- Project repository: github.com/xuexingzxx/PDE-Formalization

SKILLS

Mathematics: Probability, Stochastic Analysis, Partial Differential Equations, Random Dynamical Systems, Sequential Monte Carlo / Particle Filtering, Numerical Analysis

Programming: Python, PyTorch

Additional Tools: Lean (Theorem Proving / Formal Verification), MATLAB, C++, SQL

Languages: English, Mandarin Chinese

HONORS & TEACHING

- National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) — 2019
- Teaching Assistant, College Algebra — Oklahoma State University (Spring 2026 – Present)